

Metrics and Evaluation Guide

Regression, GLM and classification metrics explained simply with business meaning

Ruturaj Mokashi

Why metrics depend on the task

A metric is not just a number. It tells you what kind of mistake the model is being judged on. Regression, count modeling and classification need different metrics because their targets behave differently.

Metric	Used for	Plain meaning	Notebook
MSE	Regression	Average squared error. Penalizes large mistakes strongly.	1
RMSE	Regression	Square root of MSE; error in target units.	1, 2, 3
MAE	Regression/count	Average absolute mistake; easy for business users.	1, 3
R2	Regression	Share of variance explained by the model.	1, 2, 3
Adjusted R2	Regression	R2 adjusted for number of features.	1
Mean Poisson Deviance	Count models	Count-aware error for Poisson-style predictions.	3
AIC/BIC	Statistical models	Fit-vs-complexity model comparison tools.	3, 5
Accuracy	Classification	Share of correct predictions.	4, 5
Precision	Classification	Of predicted positives, how many were truly positive.	4
Recall	Classification	Of actual positives, how many were found.	4
F1	Classification	Balance between precision and recall.	4
ROC-AUC	Classification ranking	How well probabilities rank positives above negatives.	4
PR-AUC	Imbalanced classification	Precision-recall quality across thresholds.	4
Log loss	Probability classification	Penalizes overconfident wrong probabilities.	4
Calibration	Probability trust	Checks if predicted probabilities match actual outcomes.	4
Lift	Marketing prioritization	Shows business value of targeting top-scored groups.	4

Metric selection rules

Question	Best metric starting point
How far are numeric predictions from actual values?	MAE and RMSE
How much variation is explained?	R2 and adjusted R2
Is a count model better than another count model?	Mean Poisson Deviance and AIC/BIC
Is a classifier good at ranking customers?	ROC-AUC and PR-AUC
Should we target customers?	Lift, threshold trade-off, precision and recall

References and source foundation

The documents use the completed Day 12 notebooks as the main project evidence. The theory explanations are also aligned with the uploaded course materials and official library documentation.

Source	How it was used
Uploaded course guides: leit03.pdf and Haupt_leitfaden03.pdf	OLS, least squares, residuals, regression line, correlation, interpolation/extrapolation.
Uploaded ML slides: Methode der Kleinsten Quadrate and Typen des Machinellen Lernens	Supervised learning, regression vs classification, gradient descent, logistic regression context.
scikit-learn Linear Models User Guide - https://scikit-learn.org/stable/modules/linear_model.html	Linear Regression, Ridge, Lasso, ElasticNet, Logistic Regression, PoissonRegressor.
scikit-learn Model Evaluation User Guide - https://scikit-learn.org/stable/modules/model_evaluation.html	MSE, MAE, R2, ROC-AUC, PR-AUC, precision, recall, F1, calibration.
statsmodels GLM documentation - https://www.statsmodels.org/stable/glm.html	Generalized Linear Models, Poisson family and model interpretation.
statsmodels NegativeBinomial documentation - https://www.statsmodels.org/stable/generated/statsmodels.discrete.discrete_model.NegativeBinomial.html	Negative Binomial implementation details and count-model context.
UCLA IDRE Negative Binomial Regression notes - https://stats.oarc.ucla.edu/r/dae/negative-binomial-regression/	Practical explanation of Negative Binomial as a model for overdispersed count outcomes.